

PROJECT.md — Summer Kids App / Personalized Learning Platform

Read this first. This file is the entry point for any AI or human developer. It explains what this project is, where every kind of information lives, and the rules that override everything else.

Documentation map — where information lives (never duplicate)

Topic	Single source of truth
Vision, priorities, tech stack, ecosystem	PROJECT.md (this file)
Game mechanics, economy, XP, modes, UX	GAME_DESIGN.md
Learning rules, curriculum, hints, feedback, quality gates	EDUCATION.md
Alex, Katya, family, personalization, privacy of personal data	CHARACTERS.md
Architecture, code map, workflow, testing, deployment, backend/privacy/cost rules	DEVELOPMENT_RULES.md
Roadmap and every planned feature (Now/Next/Later/Maybe)	FUTURE_IDEAS.md

When a new idea or fact appears, put it in exactly one file per this table and, if needed, add a one-line cross-reference from other files. Do not restate content.

What this is

A gamified, personalized learning web app built by Yulia (Toronto) **for her two children only** — Alex (entering Grade 8, Sept 2026) and Katya (entering Grade 5, Sept 2026). Today it is a math app ("Summer Math Battle Tutor"); the long-term vision is a full personalized learning platform covering the Ontario curriculum across many subjects. Full child profiles: see CHARACTERS.md.

Vision

An outstanding private AI tutor in game form: children who genuinely enjoy learning and become independent, confident learners. Educational quality, engagement, and long-term motivation matter more than commercial polish or release schedules.

Priorities (in order — feature-driven, no fixed dates)

1. **Now — Best personalized learning platform for Alex and Katya.** Not complete. Everything else waits on this.
2. **Next — Expand beyond math** to a complete Ontario-curriculum platform. Subject order and quality gates: see EDUCATION.md → “Subject roadmap” and “Quality standards”. A new subject is added only when the previous one meets the quality bar.
3. **Later — Complete Parent Dashboard:** progress by child, strengths/gaps, custom missions and rewards, worksheet/test uploads, generated practice, report-card goals, analytics, AI recommendations, multi-child plans.
4. **Maybe — Commercial product for other families,** keeping personalization as the biggest advantage. Requires the privacy conversion described in CHARACTERS.md → “Privacy rules”.

Two-app ecosystem (hard boundary)

- **This app (learning)** = education only.
- **“Just Do It and Be Done” (separate repo/app)** = chores, routines, habits, screen time, responsibility, family management.
- **Never merge them.** They should feel like two games in one ecosystem.
- Eventually integrate through a **shared family account** exchanging achievements, XP, coins, streaks, and rewards (tracked in FUTURE_IDEAS.md).

Core personalization model

The platform personalizes learning per child by: current grade, target grade, age (auto-calculated from birthday — see CHARACTERS.md), subject, skill level, strengths and learning gaps, parent priorities, individual interests and motivation, and automatic progression over time. Grade level advances each school year and is manually adjustable by the parent. **“Summer Bridge” is one optional mode, not the core concept.**

Gameplay philosophy (summary — details in

GAME_DESIGN.md)

Responsibilities remain — effort earns privileges.

Rewards are privileges, experiences, and positive motivation — never exemptions from practice, chores, or responsibilities. No punishment mechanics, no losing earned progress, no device-removal threats. Kids request rewards; a parent approves every redemption.

Educational objectives (summary — details in EDUCATION.md)

Ontario curriculum alignment; adaptive difficulty; mastery tracking; mistakes treated as learning (3-attempt teaching flow); effort and persistence rewarded over speed; the app explains *why*, not just the answer.

Technical stack (summary — details in DEVELOPMENT_RULES.md)

- **Current MVP:** single-file `index.html` (vanilla JS + CSS inline), Vite for dev/build, localStorage state, deployed on Vercel. Neon-arcade dark aesthetic (Orbitron headings). Tablet-first.
- **Declared long-term target:** React + TypeScript, migrated gradually feature-by-feature with **zero visual or behavioral regression**. The current app is the MVP and design reference, not disposable legacy code.
- **Later backend:** family account, cloud DB, file storage, server-side AI API. Rules and provider guidance: DEVELOPMENT_RULES.md → “Backend path”.

Non-negotiable ground rules for any AI developer

1. `main` must always stay stable and usable by the children.
2. Never erase or corrupt existing child progress. Migrations need a data plan.
3. Preserve the current look, features, content, and behavior unless explicitly asked to change them.
4. Children's personal data stays private (CHARACTERS.md → Privacy rules; DEVELOPMENT_RULES.md → Data & privacy rules).
5. Follow the wrong-answer teaching flow and motivation rules in EDUCATION.md for every learning feature.

6. File new ideas in FUTURE_IDEAS.md under the correct priority bucket instead of building them immediately.